poolside/Laguna-S-2.1-FP8

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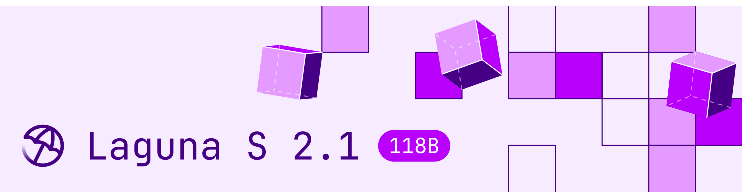
Model card · Files xet · Community 2



Use on OpenRouter · **Use on Vercel AI Gateway** · **Release blog post**

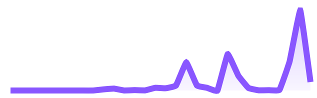
Laguna S 2.1-FP8

Laguna S 2.1-FP8 is a 117.6B total parameter Mixture-of-Experts model with 8.5B activated parameters per token designed for agentic coding and long-horizon work on a local machine. It uses Sliding Window Attention with per-head gating in 36 out of 48 layers for fast inference and low KV cache requirements.

Highlights

- **Mixed SWA and global attention layout**: Laguna S 2.1 uses softplus gating with per-layer rotary scales, enabling mixed SWA (Sliding

Downloads last month
21,664



Safetensors

Model size	118B params
Tensor type	F32 · BF16 · F8_E4M3

Chat template · Files info

Inference Providers NEW

Text Generation

This model isn't deployed by any Inference Provider.

Ask for provider support

Model tree for poolside/Laguna-S...

Base model ... poolside/Laguna-S-2.1

Quantized (56) ... this model

Finetunes ... 1 model

Collection including poolside/Lag...

Laguna S 2.1 Collection

Window Attention) and global attention layers in a 3:1 ratio (across 48 total layers)

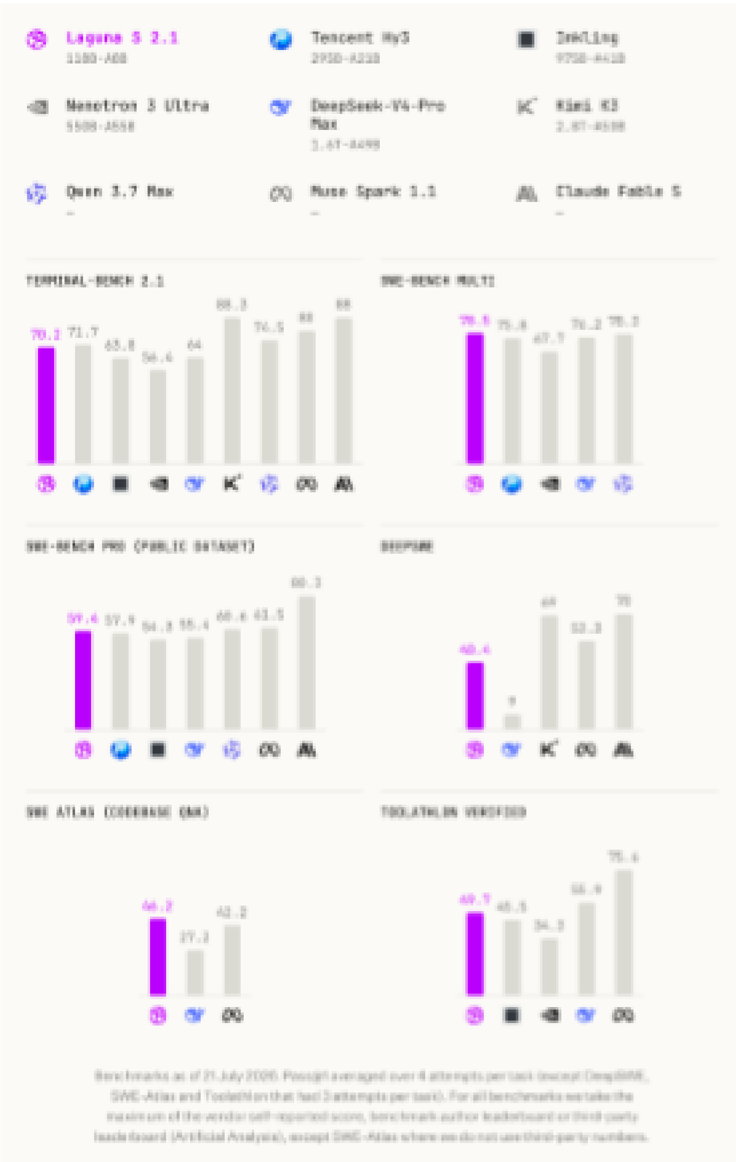
- **KV cache in FP8**: KV cache quantized to FP8, reducing memory per token
- **Native reasoning support**: Interleaved thinking between tool calls with support for enabling and disabling thinking per-request
- **Local-ready**: At 117.6B total parameters and 8.5B activated, the FP8 weights are roughly 121 GB. Available on Ollama and llama.cpp (BF16 and Q4_K_M only)
- **OpenMDW-1.1 license**: Use and modify the model and associated materials freely for commercial and non-commercial purposes (learn more about OpenMDW)

Model overview

- Training: pre-training, post-training and reinforcement learning stages
- Number of parameters: 117.6B total with 8.5B activated per token
- Optimizer: Muon
- Layers: 48 layers (12 layers with global attention, 36 layers with sliding window attention)
- Experts: 256 experts with 1 shared expert
- Sliding Window: 512 tokens
- Modality: text-to-text
- Context window: 262,144 tokens

- Reasoning support: interleaved thinking with preserved thinking

Benchmark results



Model	Size	Terminal-Bench 2.1	SWE-bench Multilingual	SWE Benc Pro (Publ Datas
Laguna S 2.1	118B-A8B	**70.2%**	**78.5%**	**59.4%**

Model	Size	Terminal-Bench 2.1	SWE-bench Multilingual	SWE Benc Pro (Publ Datas
Tencent Hy3	295B-A21B	71.7%	75.8%	57.9%
Inkling	975B-A41B	63.8%	-	54.3%
Nemotron 3 Ultra	550B-A55B	56.4%	67.7%	-
DeepSeek-V4-Pro Max	1.6T-A49B	64.0%*	76.2%	55.4%
Kimi K3	2800B-A50B	88.3%	-	-
Qwen 3.7 Max	-	74.5%*	78.3%	60.6%
Muse Spark 1.1	-	80%	-	61.5%
Claude Fable 5	-	88%	-	80.3%

Benchmarks as of 21 July 2026. Laguna S 2.1 in **bold**; a dash (-) marks a benchmark a model was not evaluated on. Scores marked * are as reported by third parties: Terminal-Bench 2.1 and DeepSWE via Artificial Analysis, SWE Atlas via Scale AI's official leaderboard, and Toolathlon Verified via its official leaderboard. Full evaluation trajectories: trajectories.poolside.ai.

Usage

Context length

This checkpoint ships configured for a 262,144-token (256K) context window. This is the configuration we recommend for best output quality.

The weights are native 1M checkpoints: training included a long-context extension stage up to 1,048,576 tokens, and quantization was calibrated at the 1M configuration. If you need more than 256K of context, restore the 1M configuration by editing `config.json`:

```
"rope_parameters": {
  "full_attention": {
    "factor": 128.0,
    "attention_factor": 1.4852030263919618
  }
},
"max_position_embeddings": 1048576
```

You may experience quality degradation with the 1M configuration. If you use it, we recommend sampling with `temperature <= 0.7` and `top_p <= 0.95`.

Recommended sampling

For the best balance of quality and reliability we recommend sampling with `temperature 0.7` and `top_p 0.95`.

Local deployment

Laguna S 2.1-FP8 is supported in vLLM, SGLang and Transformers, and TRT-LLM thanks to the support of the team at NVIDIA. Use Laguna S 2.1 with Ollama (with MLX support) or Llama.cpp (BF16 and Q4_K_M only) for the best results on your local machine.

vLLM

> *"Serving requires vLLM 0.25.0 or later. See the main Laguna S 2.1 model card for the full recipe."*

The full vLLM recipe is on the main Laguna S 2.1 model card and on the vLLM recipes page. Quantization is detected automatically from `quantization_config` in this checkpoint, so the same command works with `poolside/Laguna-S-2.1-FP8` substituted for the model ID. Set `VLLM_BLOCKSCALE_FP8_GEMM_FLASHINFER=0` when serving with vLLM.

> **Optional: speculative decoding with DFlash.** Pair with the quantization-matched draft model poolside/Laguna-S-2.1-DFlash-FP8 by adding `--speculative-config '{"model":"poolside/Laguna-S-2.1-DFlash-FP8","num_speculative_tokens":15,"method":"dflash"}'` to the serve command.

SGLang

Laguna S 2.1 is supported in SGLang via sgl-project/sglang#24204. Quantization is detected automatically from `quantization_config`, so no extra flags are required. See the SGLang cookbook entry and the main Laguna S 2.1 model card for a serving recipe.

Transformers

The full Transformers recipe is on the main Laguna S 2.1 model card. Substitute `poolside/Laguna-S-2.1-FP8` for the model ID; quantization is detected automatically from `quantization_config`.

TRT-LLM

Laguna S 2.1 support ships in TensorRT-LLM `>=1.3.0rc16`; see the install recipe on the main Laguna S 2.1 model card. Substitute `poolside/Laguna-S-2.1-FP8` for the model ID; quantization is detected automatically from `quantization_config`, no extra flags required.

```
from tensorrt_llm import LLM

llm = LLM(model="poolside/Laguna-S-2.1-FP8",
```

llama.cpp

Serve the GGUF builds with Poolside's llama.cpp fork, branch laguna. See the main Laguna S 2.1 model card for the build recipe. llama.cpp serves the BF16, Q8_0, and Q4_K_M GGUF builds, not the FP8 safetensors.

Ollama

Available on the Ollama library.

Controlling reasoning

Laguna S 2.1-FP8 uses the same reasoning controls (interleaved thinking, preserved reasoning, and the `enable_thinking` flag) as the base model. See the Controlling reasoning section of the main Laguna S 2.1 model card.

License

This model is licensed under the OpenMDW-1.1 License.

Intended and Responsible Use

Laguna S 2.1-FP8 is designed for software engineering and agentic coding use cases, and you are responsible for confirming that it is appropriate for your intended application. Laguna S 2.1-FP8 is subject to the OpenMDW-1.1 License, and should be used consistently with Poolside's Acceptable Use Policy. We advise against circumventing Laguna S 2.1-FP8 safety guardrails without implementing substantially equivalent mitigations appropriate

Please report security vulnerabilities or safety concerns to security@poolside.ai.